

HYT ENGINEERING COMPANY PVT. LTD.

Product Catalogue

CNC wheel lathes · Wheel & axle shop machines · Depot equipment
Turnkey railway workshops · Lifetime service

SAMPLE CATALOGUE — FOR DESIGN AND TESTING

Illustrations are generated sample images. Specifications are placeholders unless confirmed by HYT in a technical offer.

COMPANY INTRODUCTION

Precision that keeps fleets moving

HYT Engineering designs, builds, installs and maintains machines for railway wheel and axle work. The company traces its origins to 1977, when it began building small drilling machines and lathes in Pune, India, and was incorporated as HYT Engineering Company Pvt. Ltd. in 1989.

Today the portfolio covers CNC underfloor, tandem, surface and portal wheel lathes; vertical turning and axle machining; wheel presses, measuring and balancing machines; depot equipment; and complete turnkey wheel shops – supported by installation, maintenance contracts, spare parts, retro-fitment and training.



1977
Origins of the company

1,100+*
Machines delivered (company figure)

ISO
9001:2015 · 14001:2015 · 45001:2018

Pune
Design and manufacturing, India

Product families in this catalogue

Wheel Lathes – CNC Surface Wheel Lathe, CNC Underfloor Wheel Lathe (YUL-08), CNC Underfloor Wheel Lathe (YUGL), CNC Tandem Underfloor Wheel Lathe (YUL-12T), CNC Tandem Underfloor Wheel Lathe (YUGLX), CNC Portal Wheel Lathe, CNC Roller Type Portal Wheel Lathe

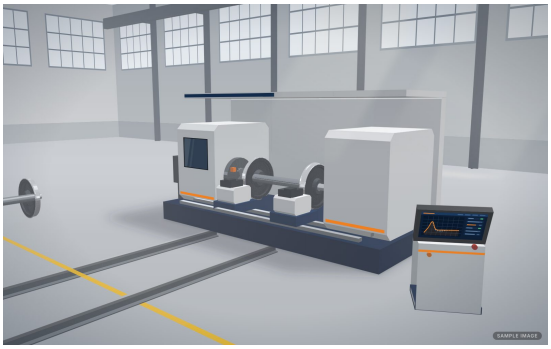
Wheel & Axle Shop Machines – CNC Vertical Turning & Boring Machine, CNC Axle Journal Turning & Burnishing Lathe, CNC Axle Turning Lathe, Wheel Press, Wheel Measuring Station, Dynamic Wheel Balancing Machine

Depot & Workshop Equipment – Multipurpose Armature Machine, Automated Train Washing Plant, Rail cum Road Vehicle (Shunter)

Partner Products – SKC Gleittechnik (Germany)

* Figure to be confirmed by HYT before publication.

WHEEL LATHES



CNC Surface Wheel Lathe

Model: YSL-08 / YSL-12 / YSL-21

Floor-level CNC lathe for re-profiling removed wheelsets and brake discs, available as models YSL-08, YSL-12 and YSL-21.

Parameter	YSL-08	YSL-12	YSL-21
Machine type	CNC surface wheel lathe	CNC surface wheel lathe	CNC surface wheel lathe
Tread diameter, maximum	1250 mm	On request	On request
Tread diameter, minimum	On request*	On request	On request
Wheel width	75–145 mm	On request	On request
Axle length	1200–2750 mm*	On request	On request
Maximum wheelset weight	50 kN	On request	On request
Brake-disc machining diameter	250–700 mm	On request	On request

Specification placeholder – values marked “On request” or * are confirmed in HYT’s technical offer.

WHEEL LATHES



CNC Underfloor Wheel Lathe (YUL-08)

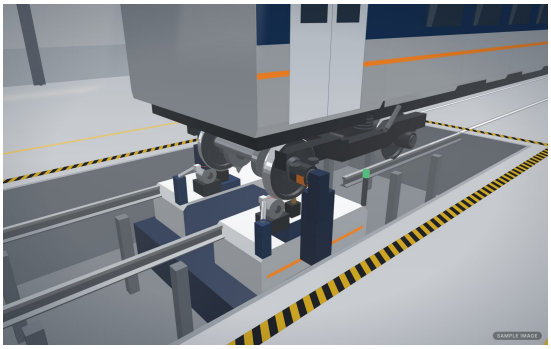
Model: YUL-08

Single-axle pit lathe that re-profiles both wheels of a wheelset in the vehicle, without dropping bogies or removing brake gear.

Parameter	YUL-08
Machine type	CNC underfloor (pit) wheel lathe, single axle
Main drive motor capacity	4 × 15 kW
Wheels machined simultaneously	2 (one wheelset)
Tread diameter range	On request*
Track gauge	Broad, standard or metre gauge – to customer specification*
Maximum axle load	On request
Profiles	Customer-specified wheel profiles stored in the CNC

Specification placeholder – values marked “On request” or * are confirmed in HYT’s technical offer.

WHEEL LATHES



CNC Underfloor Wheel Lathe (YUGL)

Model: YUGL

Next-generation underfloor wheel lathe for in-vehicle re-profiling, positioned for depots that want a higher degree of automation.

Parameter	YUGL
Machine type	CNC underfloor (pit) wheel lathe, single axle
Wheels machined simultaneously	2 (one wheelset)
Main drive	On request
Tread diameter range	On request*
Track gauge	To customer specification*
Maximum axle load	On request
Measuring system	On request

Specification placeholder – values marked “On request” or * are confirmed in HYT’s technical offer.

WHEEL LATHES



CNC Tandem Underfloor Wheel Lathe (YUL-12T)

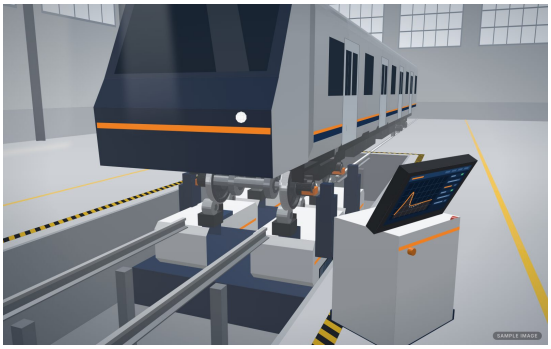
Model: YUL-12T

Two synchronised underfloor lathes that re-profile both wheelsets of a bogie in one setting, cutting re-profiling time per vehicle.

Parameter	YUL-12T
Machine type	CNC tandem underfloor (pit) wheel lathe – two synchronised machines
Configuration	One fixed machine, one traversing machine to suit bogie wheelbase
Wheelsets machined per setting	2 (one bogie)
Main drive motor capacity	4 × 15 kW
Tread diameter range	540–1500 mm
Cutting speed	20–120 m/min
Bogie wheelbase range	On request*

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WHEEL LATHES



CNC Tandem Underfloor Wheel Lathe (YUGLX)

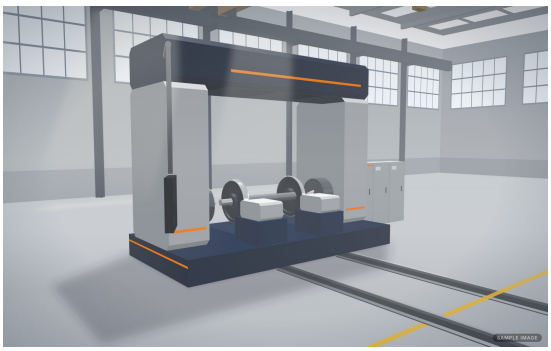
Model: YUGLX

Next-generation tandem underfloor lathe that re-profiles a complete bogie in the vehicle, positioned for higher automation.

Parameter	YUGLX
Machine type	CNC tandem underfloor (pit) wheel lathe
Wheelsets machined per setting	2 (one bogie)
Main drive	On request
Tread diameter range	On request*
Bogie wheelbase range	On request*
Track gauge	To customer specification*
Maximum axle load	On request

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WHEEL LATHES



CNC Portal Wheel Lathe

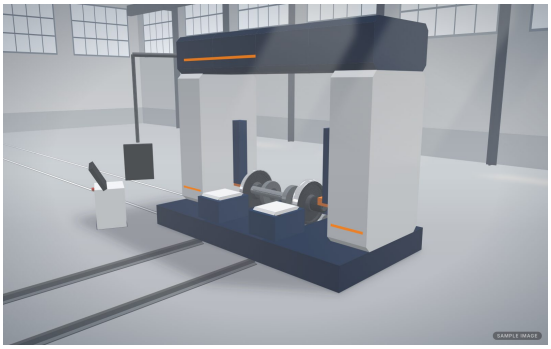
Model: YPL-02

Rigid portal lathe for re-profiling removed wheelsets in wheel and axle shops, with a 2 × 55 kW main drive for heavy cuts.

Parameter	YPL-02
Machine type	CNC portal wheel lathe for removed wheelsets
Tread diameter, maximum	1250 mm
Tread diameter, minimum	750 mm*
Wheel width	75–145 mm
Axle length	1850–2750 mm*
Maximum wheelset weight	50 kN
Main drive	2 × 55 kW

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WHEEL LATHES



CNC Roller Type Portal Wheel Lathe

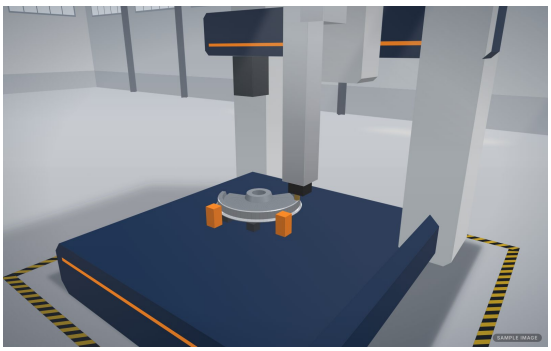
Model: YPL-04

Roller-type portal wheel lathe for re-profiling removed wheelsets, complementing the YPL-02 in wheel shops and overhaul workshops.

Parameter	YPL-04
Machine type	CNC roller type portal wheel lathe for removed wheelsets
Tread diameter range	On request*
Wheel width	On request
Axle length	On request*
Maximum wheelset weight	On request
Main drive	On request
Cutting speed	On request

Specification placeholder – values marked “On request” or * are confirmed in HYT’s technical offer.

WHEEL & AXLE SHOP MACHINES



CNC Vertical Turning & Boring Machine

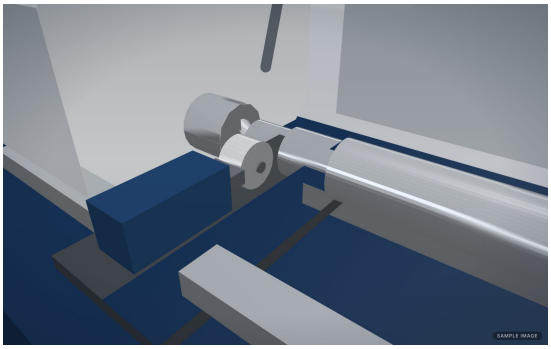
Model: YVM-04

Vertical turning and boring machine for loose railway wheels and tyres – profile, rim faces and hub bore – in new production and overhaul.

Parameter	YVM-04
Machine type	CNC vertical turning and boring machine
Typical workpieces	Railway wheels, tyres and similar round components
Table diameter	On request
Maximum turning diameter	On request
Maximum workpiece height	On request
Maximum workpiece weight	On request
Main drive	On request

Specification placeholder – values marked “On request” or * are confirmed in HYT’s technical offer.

WHEEL & AXLE SHOP MACHINES



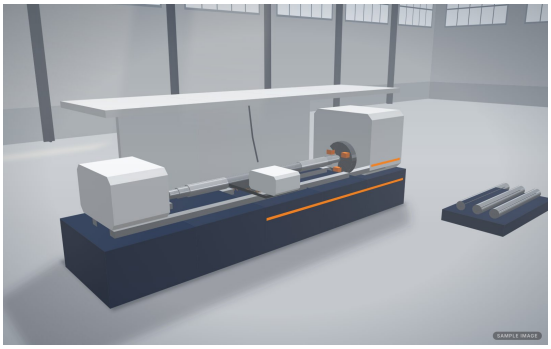
CNC Axle Journal Turning & Burnishing Lathe

Restores and burnishes axle journals during wheelset overhaul, for accurate, durable bearing seats.

Parameter	Value
Machine type	CNC lathe for axle journal turning and burnishing
Typical work	Restoration of bearing journals and adjacent seats during overhaul
Maximum axle length	On request*
Journal diameter range	On request
Maximum workpiece weight	On request
Burnishing tool	On request
Main drive	On request

Specification placeholder – values marked “On request” or * are confirmed in HYT’s technical offer.

WHEEL & AXLE SHOP MACHINES



CNC Axle Turning Lathe

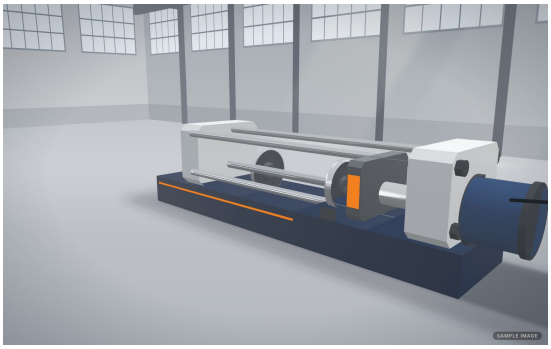
Model: ATL-01

Turns complete railway axles from raw or forged blanks to the finished contour for new wheelset production.

Parameter	ATL-01
Machine type	CNC axle turning lathe
Workpiece	Raw or forged railway axles
Maximum axle length	On request
Maximum turning diameter	On request
Maximum axle weight	On request
Main drive	On request
Workpiece clamping	On request

Specification placeholder – values marked “On request” or * are confirmed in HYT’s technical offer.

WHEEL & AXLE SHOP MACHINES



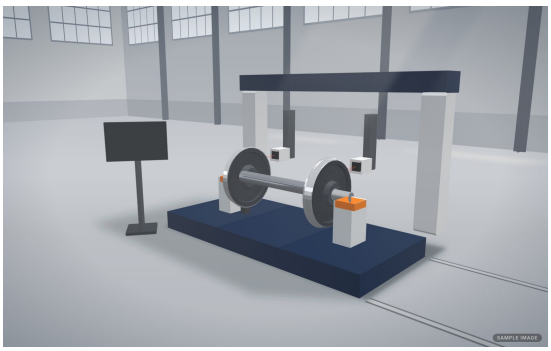
Wheel Press

Hydraulic presses for mounting wheels on axles and removing them during overhaul, with controlled, documented press-fits.

Parameter	Value
Press type	Hydraulic wheel press for mounting and/or dismounting
Maximum press force	On request
Maximum axle length	On request*
Wheel diameter range	On request
Back-to-back setting	To customer specification*
Press-fit recording	On request
Wheelset handling	On request

Specification placeholder – values marked “On request” or * are confirmed in HYT’s technical offer.

WHEEL & AXLE SHOP MACHINES



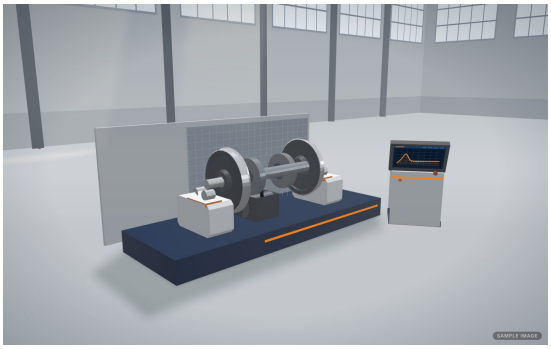
Wheel Measuring Station

Automatic measurement of wheel and wheelset geometry before and after machining, with records that support decisions and traceability.

Parameter	Value
Measured parameters	Tread diameters, flange height, flange thickness, back-to-back distance – further parameters on request
Measuring principle	On request
Measuring accuracy	On request
Cycle time	On request
Wheel diameter range	On request*
Track gauge	To customer specification*
Data output	On request

Specification placeholder – values marked “On request” or * are confirmed in HYT’s technical offer.

WHEEL & AXLE SHOP MACHINES



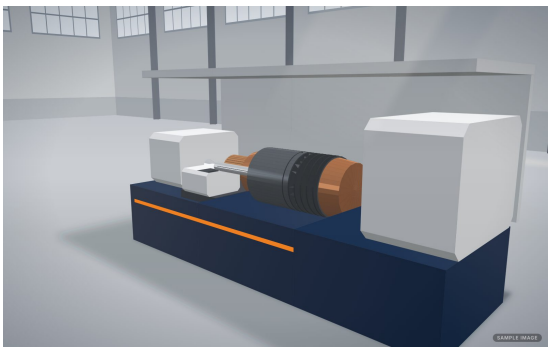
Dynamic Wheel Balancing Machine

Measures the dynamic unbalance of railway wheels and wheelsets so it can be corrected before service – for smoother running and longer component life.

Parameter	Value
Workpiece	Railway wheels or wheelsets – configuration on request
Balancing planes	On request
Maximum workpiece weight	On request
Wheel diameter range	On request
Balancing speed	On request
Measuring resolution	On request
Data output	On request

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DEPOT & WORKSHOP EQUIPMENT



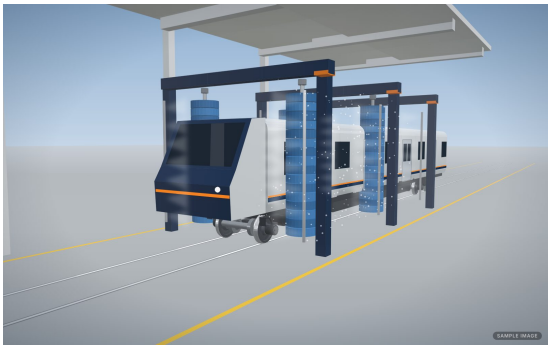
Multipurpose Armature Machine

Combines commutator turning, undercutting and related armature operations for traction-motor overhaul on one machine.

Parameter	Value
Workpiece	Traction-motor armatures
Operations	Commutator turning, mica undercutting and related operations – scope on request
Maximum armature weight	On request
Maximum armature length	On request
Commutator diameter range	On request
Drive	On request

Specification placeholder – values marked “On request” or * are confirmed in HYT’s technical offer.

DEPOT & WORKSHOP EQUIPMENT



Automated Train Washing Plant

Drive-through plant that washes train exteriors automatically, for consistent cleaning with less labour and controlled water use.

Parameter	Value
Washing type	Drive-through exterior washing
Vehicle types	To customer specification
Train passage speed	On request
Brush arrangement	On request
Water consumption per train	On request
Water recovery and treatment	On request
Detergent dosing	On request

Specification placeholder – values marked “On request” or * are confirmed in HYT’s technical offer.

DEPOT & WORKSHOP EQUIPMENT



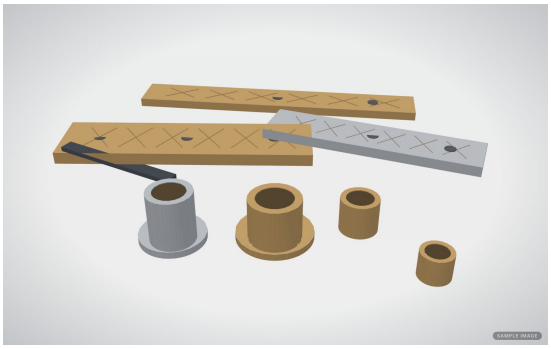
Rail cum Road Vehicle (Shunter)

Dual-mode shunter that moves rolling stock on rails and drives on roads between tracks – an alternative to locomotive shunting in depots.

Parameter	Value
Operating modes	Rail and road
Track gauge	To customer specification*
Hauling capacity	On request
Tractive effort	On request
Power unit	On request
Coupling system	To customer specification
Brake connection to hauled stock	On request

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PARTNER PRODUCTS



SKC Gleittechnik (Germany)

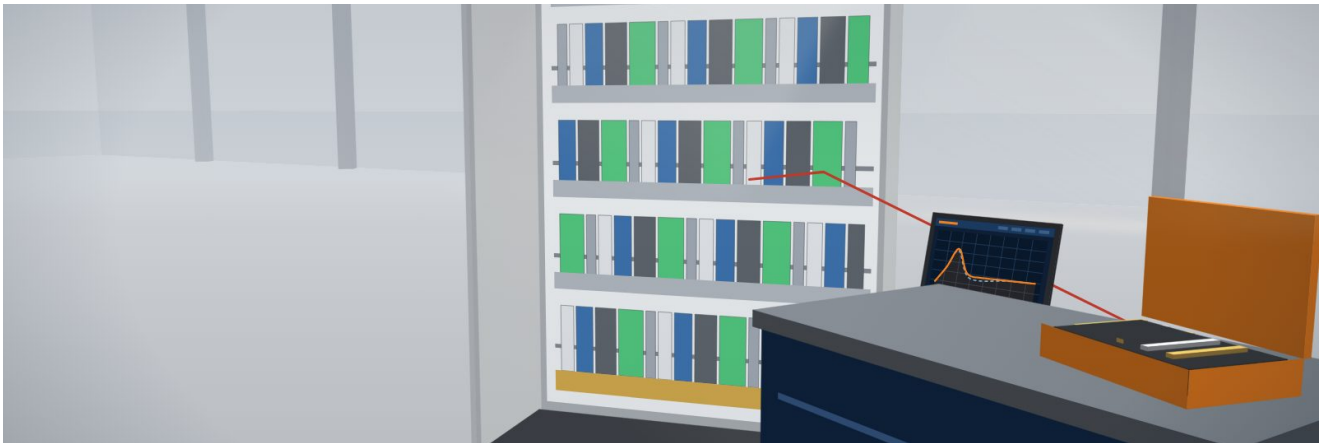
Sliding and slideway technology from SKC Gleittechnik GmbH, Germany – distributed exclusively in India by HYT.

Parameter	Details
Manufacturer	SKC Gleittechnik GmbH, Germany
Distribution in India	HYT Engineering Company Pvt. Ltd. – exclusive distributor
Product range	On request
Technical data	On request
Application advice	Through HYT
Delivery time	On request

Specification placeholder – values marked “On request” or * are confirmed in HYT's technical offer.

Support for the whole machine life

- **Installation & commissioning** – Foundation checks, alignment, acceptance trials, documented handover.
- **AMC & CMC contracts** – Planned preventive maintenance with defined scope and response.
- **Spare parts & support** – Critical spares, tooling and consumables identified and supplied.
- **Retro-fitment & upgradation** – New CNC systems, drives and measuring technology for older machines.
- **Training** – Operators, programmers and maintenance teams trained on site.
- **MAAS – wet leasing** – HYT supplies machine and crew; the customer pays for output.
- **Turnkey (EPC) projects** – Complete wheel shops and depots, from layout to lifetime support.



Contact (sample – verify before printing)

HYT Engineering Company Pvt. Ltd.
29/10, 29/10/1, 29/10/2, D-II Block
MIDC Chinchwad
Pune 411019, Maharashtra, India
Tel. +91 20 6634 3900 / +91 20 6634 3906 · Toll-free 1800 2100 456
info@hytengineering.com · https://hytengineering.com